

(1) Find the equation of the tangent plane of the given surface at the specified point.

- (i) $z = \sqrt{4 - x^2 - 2y^2}$ at $p = (1, -1, 1)$
- (ii) $z = y \log x$ at $p = (1, 4, 0)$

(2) Find the differentials of the following functions

- (i) $z = x^3 \log(y^2)$
- (ii) $w = \log \sqrt{x^2 + y^2 + z^2}$

(3) Use the chain rule to find dz/dt and dw/dt for

- (i) $z = \sin x \cos y$ where $x = \pi t$, $y = \sqrt{t}$
- (ii) $w = xy + yz^2$ where $x = e^t$, $y = e^t \sin t$, $z = e^t \cos t$

(4) Use the chain rule to find the indicated partial derivatives

- (i) $z = x^2 + xy^3$ where $x = uv^2 + w^3$, $y = u + ve^w$.
Find $\frac{\partial z}{\partial u}$, $\frac{\partial z}{\partial v}$, $\frac{\partial z}{\partial w}$ when $u = 2$, $v = 1$, $w = 0$
- (ii) $R = \log(u^2 + v^2 + w^2)$ where $u = x + 2y$, $v = 2x - y$, $w = 2xy$.
Find $\frac{\partial R}{\partial x}$, $\frac{\partial R}{\partial y}$ when $x = y = 1$

(5) Let $z = f(x, y)$.

- (i) Suppose $x = r \cos \theta$, $y = r \sin \theta$.
(a) Show that

$$\left(\frac{\partial z}{\partial x}\right)^2 + \left(\frac{\partial z}{\partial y}\right)^2 = \left(\frac{\partial z}{\partial r}\right)^2 + \frac{1}{r^2} \left(\frac{\partial z}{\partial \theta}\right)^2$$

- (b) Show that

$$\frac{\partial^2 z}{\partial x^2} + \frac{\partial^2 z}{\partial y^2} = \frac{\partial^2 z}{\partial r^2} + \frac{1}{r^2} \frac{\partial^2 z}{\partial \theta^2} + \frac{1}{r} \frac{\partial z}{\partial r}$$

- (ii) Suppose $x = s + t$, $y = s - t$. Show that

$$\left(\frac{\partial z}{\partial x}\right)^2 - \left(\frac{\partial z}{\partial y}\right)^2 = \frac{\partial z}{\partial s} \frac{\partial z}{\partial t}$$

(6) Find the directional derivative of f at $P = [x_0, y_0, z_0]$ in the direction v .

- (i) $f(x, y) = 1 + 2x\sqrt{y}$, $P = [3, 4]$, $v = [4, -3]$
- (ii) $f(x, y, z) = x/(y + z)$, $P = [4, 1, 1]$, $v = [1, 2, 3]$

(7) Find the directional derivative of $f(x, y, z) = x^2 + y^2 + z^2$ at $P = [2, 1, 3]$ in the direction of the origin.

(8) Find the maximal and minimal rates of change of f at the given point P , and determine the direction in which this occurs.

- (i) $f(x, y, z) = x^2 y^3 z^4$ at $P = [1, 1, 1]$.
- (ii) $f(x, y, z) = \ln(xy^2 z^3)$ at $P = [1, -2, -3]$.

(9) Find equations of the tangent planes of the given surfaces at the specified points.

- (i) $x^2 + 2y^2 + 3z^2 = 21$ at $P = [4, -1, 1]$
- (ii) $x - z = 4 \arctan(yz)$ at $P = [1 + \pi, 1, 1]$.

(10) Find the points on the ellipsoid $x^2 + 2y^2 + 3z^2 = 1$ at which the tangent plane is parallel to the plane $3x - y + 3z = 1$.